Anees ur Rehman

araneeskhan | araneeskhan | araneeskhan.vercel.app | aneesurrehman1358@gmail.com
Paris, France — +33 7 80 80 07 92

Summary

Software Engineer and AI Master's student at Cnam, Paris, seeking to pursue a PhD focused on applying Artificial Intelligence to Software Engineering and building secure, trustworthy software systems. Strong background in full-stack development, Large Language Models (LLMs), and scalable architecture, with hands-on experience building and publishing an Android game on Google Play.

Published Mobile Game

Business Empire: Underworld Google Play

- **Built and deployed an Android business simulation and strategy game on Google Play**; initially released September 23, 2026.
- Developed the application with **React Native, Expo, TypeScript, Zustand, and Supabase**, spanning gameplay logic, mobile UI, backend integration, testing, and Android publishing.
- Implemented interconnected property upgrades, investment markets, industrial production, transport management, research, and long-term economic progression.
- Built offline earnings, versioned local persistence and save migrations, guest play, and authenticated cloud-save workflows; integrated optional rewarded ads through Unity LevelPlay.

Research Interests

- **AI for Software Engineering:** Integrating AI-driven methodologies into the software development lifecycle to automate rigorous testing, code generation, and vulnerability detection. Exploring novel paradigms that bridge machine learning models with traditional engineering workflows to fundamentally enhance system reliability and developer productivity.
- **LLMs and AI Agents:** Leveraging Large Language Models and autonomous agentic systems for complex code analysis, dynamic refactoring, and intelligent software evolution. Investigating the deployment of multi-agent architectures to solve intricate, system-level design challenges and optimize continuous integration pipelines.

Achievements

- **1st Position, Final Year Project (FYP):** Awarded 1st place at the COMSATS Career Expo 2024 for the "Campus Sports Sphere" capstone project.

Publications

Evaluating Retrieval-Augmented Generation Strategies for Repository-Level Code Understanding

- **Published:** August 2026 | **Platform:** Zenodo Preprint (Indexed in OpenAIRE)
- **DOI:** 10.5281/zenodo.21813119
- Evaluated dense, sparse (BM25), and hybrid Reciprocal Rank Fusion (RRF) retrieval strategies across Python, Java, and C++ codebases using an LLM-as-a-Judge evaluation framework.

Education

Master of Artificial Intelligence for Connected Industries 2025 - present

Conservatoire National des Arts et Métiers (Cnam), Paris

- **Academic Excellence (M1):** High Distinction equivalent. Achieved a perfect score (20/20) in Operating Systems & Computer Architecture, and outstanding marks in Data Management (18.7/20) and AI & Machine Learning (15.8/20).
- **Relevant Coursework:** Operating Systems & Computer Architecture, AI & Machine Learning, Distributed & Federated Learning, Performance Evaluation.

Bachelor of Science in Computer Science 2021 - 2025
COMSATS University Islamabad

Academic Research

Year-Long Research Project: DNS Cache Poisoning Attack Cnam, Paris
- Analyzed side-channel vulnerabilities in the Linux kernel's ICMP error processing that bypass Source Port Randomization.
- Proposed a novel "0x20 + TCP Hybrid Defense" mechanism utilizing randomized capitalization and preemptive TCP switching to deterministically neutralize cache poisoning threats without disrupting Path MTU Discovery.
- **AI Integration:** Extending the defense framework with Machine Learning models for real-time anomaly detection to dynamically identify and block complex side-channel exploitation patterns.

Work Experience

Software Engineer, Sybrid IT Solution (remote) Dec 2024 - August 2025
- Built and deployed full-stack applications using Next.js, React, React Native, and Node.js with SQL, Firebase, and MongoDB backends.
- Delivered production releases on schedule within cross-functional teams; contributed to architecture discussions for AI/ML data integration.

Software Engineer Intern, Student Startup Business Center (SSBC) June 2024 - Aug 2024
- Developed scalable web apps using Node.js, Express.js, React.js, and MongoDB; built RESTful APIs and reusable front-end components.

Key AI & Software Engineering Projects

Campus Sports Sphere — AI-Driven Sports Management System Source
- Engineered a full-stack platform (React Native/Node.js) featuring an intelligent recommendation engine for equipment and ground reservations based on user history and real-time weather data.
- Developed an AI module using TensorFlow.js for image recognition and automatic, skill-based team formations, utilizing a Round-Robin algorithm for automated match scheduling.
- Trained and integrated a custom AI Chatbot to provide instant, contextual support for sports events and complex reservation queries.

RecoverCare — AI-Powered Health Platform Source
- Engineered a privacy-first monitoring system integrating local Natural Language Processing (NLP) for automated sentiment analysis and smart triage of patient communications.
- Developed an AI Recovery Predictor utilizing Machine Learning regression on vitals time-series data to forecast healing trajectories and preemptively detect medical anomalies.
- Architected a Dockerized Node.js/PostgreSQL backend executing all AI/ML models locally (API-free), ensuring 100% patient data sovereignty and privacy compliance.

DocuForge — Offline AI Formatting Engine Source
- Built a fully offline, AI-independent formatting engine using FastAPI, python-docx, and spaCy that standardizes raw text into publication-ready documents.
- Demonstrates applied NLP within software engineering, utilizing zero external API dependencies for enhanced reliability and privacy.

Tourism Project — Data Structure and Algorithm Source
- Engineered a C++ travel routing engine utilizing Dijkstra's Algorithm ($O(V^2)$) on directed graph adjacency matrices for optimal pathfinding.
- Architected a custom File I/O persistent state engine to serialize and flawlessly reconstruct complex, nested pointer-based linked lists across application cycles.
- Implemented rigorous memory management protocols, utilizing raw pointer manipulation and FIFO queues for safe garbage collection with zero memory leaks.

QuickBite — AI-Powered Food Delivery Application Source

– Engineered a robust, entirely on-device AI engine featuring Semantic NLP Search (TF-IDF) and a Content-Based Recommendation Engine (Jaccard Similarity) for highly personalized suggestions.
– Developed a Dynamic Prep-Time Predictor via Multiple Linear Regression and implemented Smart Cart "Apriori" Cross-Selling to heuristically boost statistically complimentary items.
– Architected a high-performance React Native frontend, integrating a simulated on-device TFLite computer vision pipeline, ensuring local processing without network inference latency without external API calls.

AI ShowCase — AI-Powered Development Platform Source

– Built a developer productivity hub with React.js, Node.js, and MongoDB; integrated LLMs (OpenAI API) for automated resume analysis and smart analytics.

Bankee — Modern Mobile Banking App Source

– Engineered a cross-platform mobile application using React Native and Expo, integrating Firebase Authentication and Firestore for secure, real-time data synchronization.
– Implemented comprehensive financial management workflows, featuring secure P2P transfers, real-time transaction history, subscription management, and push notifications.

Java Enterprise Applications — College & Hotel Management Systems Source

– Engineered object-oriented Java applications (GUI and Console) featuring role-based access control, resource allocation (courses/rooms), and robust file I/O data persistence.

Skills

AI & ML:	Machine Learning, Deep Learning, Neural Networks, Scikit-Learn, XGBoost, TensorFlow.js, NLP (spaCy, TF-IDF), Recommender Systems, LLMs, Federated Learning (Flower), Pandas
Architecture:	Data Structures & Algorithms, On-Device ML, Memory Management, REST/FastAPI, PostgreSQL, MongoDB
Languages:	Python, C++, TypeScript, JavaScript, Java, C, SQL, Bash
Full-Stack:	React.js, Next.js, React Native, Expo, Node.js, Express.js, Flask, Firebase, Supabase, Zustand, Tailwind CSS
DevOps & Tools:	Docker, Kubernetes, AWS, Git, Linux Environments

Certifications

Intermediate Machine Learning Kaggle

– Engineered and optimized robust machine learning workflows using Scikit-Learn and XGBoost on tabular datasets.
– Mastered advanced data preprocessing techniques, including handling missing values via multivariate and simple imputation, and encoding high-cardinality categorical variables using Ordinal and One-Hot Encoding.
– Implemented clean, production-ready architectures using Scikit-Learn Pipelines to prevent data leakage.

Intro to Machine Learning Kaggle

– Developed and evaluated predictive machine learning models using Scikit-Learn and Pandas.
– Built workflows encompassing exploratory data analysis, feature selection, and data splitting for validation.
– Covered the mechanics of overfitting and underfitting using Decision Tree Regressors, optimized hyperparameters through validation metrics (MAE), and implemented Random Forest Regressors to engineer robust ensemble models for predictive competitions.

Languages

French A2
English C1 (IELTS: 7.0)